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cdmetapopR: An interface for the the eco-evolutionary simulation software CDMetaPOP

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 20 October 2026
Published: unpublished

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Summary

This R package is a companion to CDMetaPOP, an individual-based, spatially explicit population genetics and demography simulation model built on Python architecture (Casey C. Day et al., 2023; Erin L. Landguth, Bearlin, et al., 2017). The package includes (1) five Shiny applications that facilitate the construction of the fundamental input files required by CDMetaPOP, (2) five functions for constructing the same fundamental input files as R objects; (3) a wrapper for launching CDMetaPOP from within R; (4) tools for importing simulation outputs as data frames for custom plotting and analysis; as well as (5) a set of functions for the visualization and analysis of simulation outputs using pre-defined workflows.

Statement of need

This package is designed to facilitate the use of CDMetaPOP by an audience comprised of researchers across diverse fields, including evolutionary biologists, landscape ecologists, geneticists, and epidemiologists. cdmetapopR provides an R-based platform for new users interested in applying CDMetaPOP to their own study systems. By offering user-friendly tools for first-time implementation, the package lowers the barrier to entry and is expected to promote broader adoption of CDMetaPOP across multiple research disciplines.

State of the field

CDMetaPOP is a well-established simulation framework in landscape ecology and population genetics that builds upon the individual-based cost-distance model CDPOP (Erin L. Landguth & Cushman, 2010) to integrate demography, genetics, spatial dynamics, temporal variability, disease transmission, and multispecies interactions (Casey C. Day et al., 2023; Erin L. Landguth, Bearlin, et al., 2017; Erin L. Landguth et al., 2026). While other spatially explicit eco-evolutionary simulators exist at varying levels of complexity, researchers in ecology and evolutionary biology increasingly rely on the R programming environment for spatial data preparation, statistical modeling, and post-simulation visualization. Bridging standalone simulation engines with R through dedicated companion packages has become standard practice in the field to streamline workflows and improve reproducibility. Previously, however, no R interface existed for CDMetaPOP; users had to manually format complex comma-separated value (CSV) parameter files, execute simulations via Python command-line interfaces, and write custom scripts to parse multi-directory outputs. `cdmetapopR` is the first and only interface available for CDMetaPOP, allowing users to configure, launch, and analyze simulations directly within R while Python runs seamlessly in the background, eliminating the need for users to

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write or execute Python code themselves.

Software design

The package provides two approaches for generating CDMetaPOP input parameter sets: interactive Shiny applications and R wrapper functions. The Shiny applications were developed to increase accessibility for new users and help them become familiar with the format and structure of the CDMetaPOP parameterization. Alternatively, the wrapper functions provide a more efficient and flexible workflow for experienced users that prefer the R coding environment.

Five Shiny applications can be used to construct the four core input files (e.g., make_runvars(); make_popvars(); Figure 1) required by CDMetaPOP, as well as an optional disease module file (make_disease()). Each Shiny app offers a streamlined approach to input-file creation by guiding users through a series of choices in a graphical user interface, referencing help from the CDMetaPOP manual when needed, and modifying standard default input files based on those selections. Once users have completed their inputs within a Shiny app, they can write the resulting comma-separated values (CSV) files locally and use them to launch CDMetaPOP.

CDMetaPOP is written in Python, and inputs are handled as CSV files. The cdmetapopR wrapper provides an accessible format for working with CDMetaPOP within the R coding environment. The wrapper provides a set of functions that can create input parameter sets and store them as R6 objects, which can then be used to launch simulations using the launch_cdmetapop() function. One function exists for each of the five primary input file types (e.g., RunVars(), PopVars()), and these can be created by reading an existing file from a local path, using default values, or by assigning values for individual parameters. The functions read_cdmetapop() and write_cdmetapop() also provide seamless reading, writing, and converting between CSV files and R objects.

The launch_cdmetapop() function opens a command prompt (Windows) or terminal (Linux/macOS) that calls Python and supplies the five arguments needed to launch CDMetaPOP simulations (Figure 1). Upon launch, the package creates a unique output directory, writes all R6 input objects as the CSV files needed to run the simulation, and returns the output directory file path that can be used to explore outputs with the summary functions. The five arguments accepted by launch_cdmetapop() are: 1) the file path to the Python executable; 2) the file path to the CDMetaPOP.py file; 3) the file path to the base input file directory; 4) the name of the corresponding RunVars file; and 5) the output directory name.

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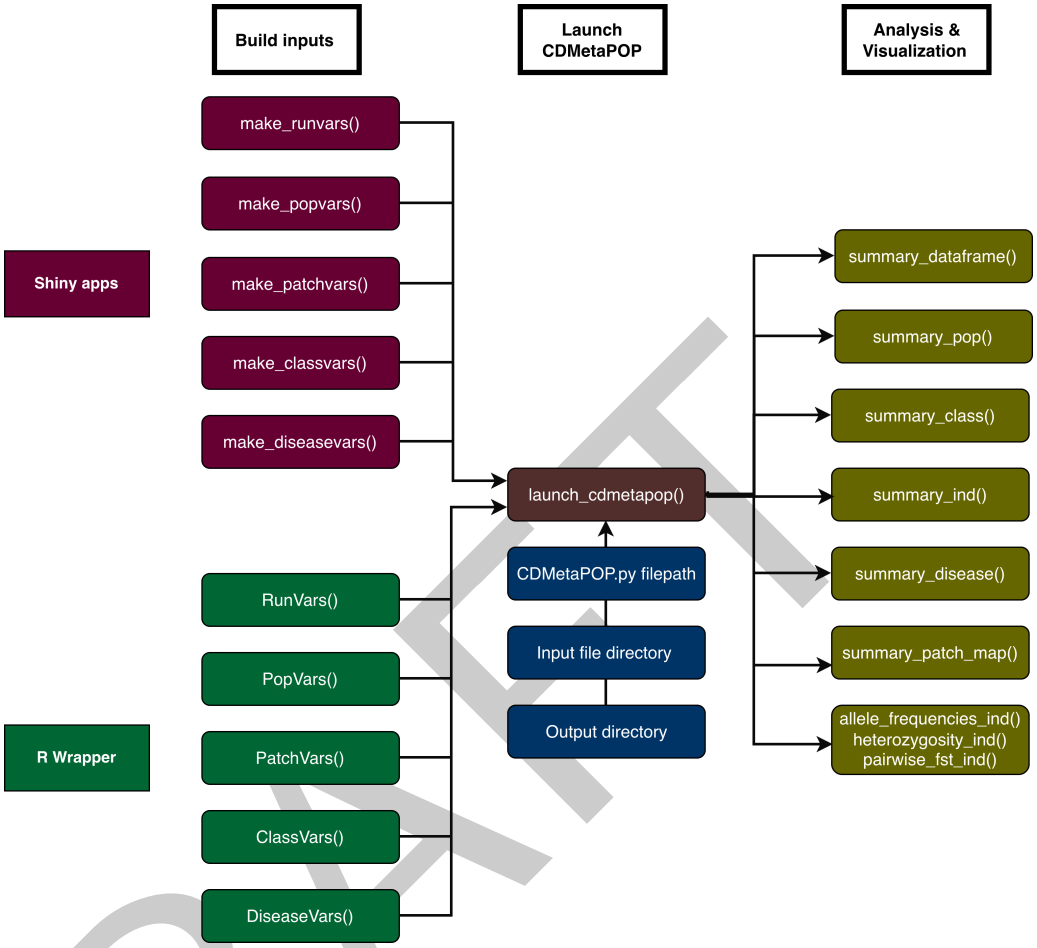


Figure 1: Figure 1: cdmetapop.R package structure.

The summary_dataframe() function converts CDMetaPOP output files into data frames, giving the user flexibility to conduct custom plotting and analyses. In addition, five classes of built-in functions can be used for visualization analysis of CDMetaPOP outputs: 1) summary_pop(), reads and summarizes population demographics extracted from the summary_popAllTime.csv output file; 2) summary_class(), reads and summarizes age and size class dynamics extracted from summary_classAllTime.csv; 3) summary_ind(), conducts analysis at the individual level from 'ind*.csv' output files (e.g., movement/ demographic, histograms of age/size structure dynamics, selection); 4) summary_disease(), analyzes disease dynamics ; and 5) summary_genetics(), summarizes population genetics and structure. These functions are designed to facilitate the analysis of CDMetaPOP output and provide users with rapid, initial insights into their simulations. They handle both in-memory and file-based workflows by accepting either a dataframe or a file path as input. They can also process outputs recursively from multiple directories (e.g., across Monte Carlo replicates or different species), extracting user-specified outcome variables and generating corresponding visualizations (Figure 2).

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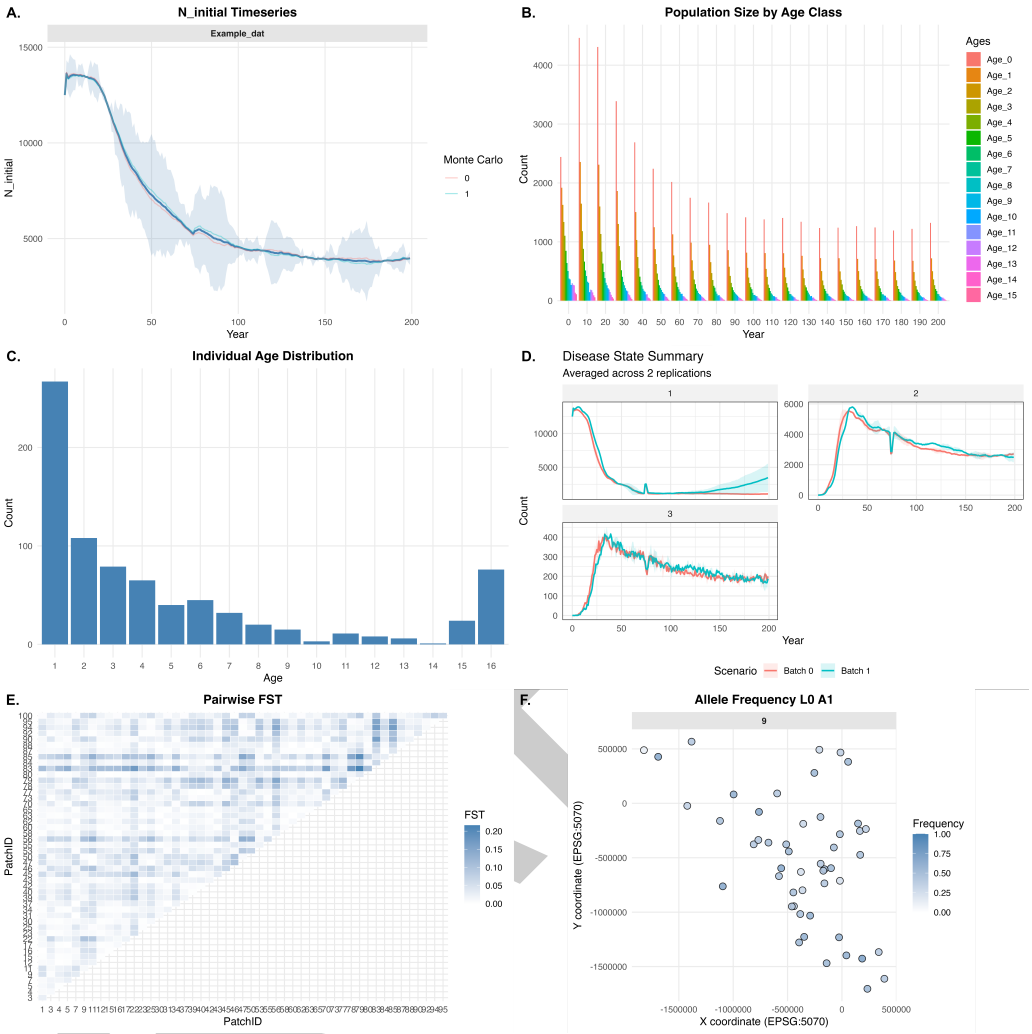


Figure 2: Figure 2: Selected visualization outputs for six of the major built-in functions of cdmetapopR. Each panel correspond to the following family of functions: A: summary_pop_n(), B: summary_class(), C: summary_ind(), D: summary_disease(), E: pairwise_fst(), F: summary_patch_map()

The package leverages an interactive user interface provided with the use of shiny (Chang et al., 2026) and shinyBS (Bailey, 2026) as well as R6 (Chang, 2025) for object-oriented class infrastructure. Additionally, several domain-standard R packages are imported for efficient data wrangling and transformation (dplyr (Wickham, François, et al., 2026) and tidyr (Wickham et al., 2025), reshape2 (Wickham, 2007), spatial (terra (Hijmans et al., 2026), graph4lg (Savary et al., 2020) and gdistance (van Etten, 2017)), population genetic analysis (adegenet (Jombart & Ahmed, 2011), poppr(Kamvar et al., 2014)), and visualization (ggplot2 (Wickham, 2016)).The package integrates with the broader R/CRAN ecosystem, utilizing roxygen2 (Wickham, Danenberg, et al., 2026) for documentation and unit testing implemented using testthat (Wickham, 2011).

Research impact statement

Spatially explicit, individual-based models (SIBMs) are powerful tools for understanding complex ecological and evolutionary systems, as they allow researchers to simulate the behavior and interactions of individual organisms across space and time. CDMetaPOP is a widely used, open-source SIBM that combines demography and genetics in a spatially explicit framework, making

it particularly useful for studying population dynamics, gene flow, selection and adaptation, disease spread, and the effects of landscape features on both real-world and hypothetical animal and plant populations [Erin L. Landguth, Bearlin, et al. (2017); Erin L. Landguth, Holden, et al. (2017); Erin L. Landguth et al. (2020); Erin L. Landguth et al. (2026); Sullivan et al. (2026); Beaman et al. (2025); Lozada et al. (2023); Bernos, Day, et al. (2023); Seaborn, Landguth, et al. (2023); Seaborn, Day, et al. (2023); Mendoza-Portillo et al. (2023); Erin L. Landguth et al. (2018); Casey C. Day et al. (2020); Casey C. Day et al. (2021); Casey C. Day et al. (2023); Casey C. Day et al. (2024); Bernos, Chang, et al. (2023); Forester et al. (2023); Escalante et al. (2018); Nathan et al. (2019); Mims et al. (2019); Menon et al. (2020); davenport2026fitness]. This package is designed to make CDMetaPOP more accessible to a wider audience of researchers, including those who may not have extensive experience with Python or command-line interfaces, or prefer working natively within the R environment. By providing user-friendly tools for input file creation and analysis of CDMetaPOP outputs within the R programming language, cdmetapopR aims to promote broader adoption of CDMetaPOP across multiple research disciplines.

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AI usage disclosure

AI tools used included: OpenAI Codex was used to assist with code generation, debugging, and documentation drafting. All AI-generated code and text were reviewed, modified, tested, and validated by the authors. The authors made all substantive design, implementation, and scientific decisions and take full responsibility for the software and manuscript content.

Acknowledgements

We thank Bill Peterman and Baylor Fain for inspiring ideas and feedback that contributed to the development of this package, and the University of Montana Computational Ecology Lab (RRID:SCR_029110) for the use of its computational resources. This work was funded by the National Science Foundation grant no. 2407848 and Seattle City Light.

References

Link to the software archive: https://doi.org/10.5281/zenodo.19687119

Bailey, E. (2026). *shinyBS: Extra twitter bootstrap components for shiny*. https://doi.org/10.32614/cran.package.shinybs

Beaman, J. E., Gates, K., Saltré, F., Hogg, C. J., Belov, K., Ashman, K., Da Silva, K. B., Beheregaray, L. B., & Bradshaw, C. J. A. (2025). A guide for developing demo-genetic models to simulate genetic rescue. *Evolutionary Applications*, *18*(5), e70092. https://doi.org/10.1111/eva.70092

Bernos, T. A., Chang, S., Giglio, R., Davenport, K., Fisher, J., Lowery, E., Bearlin, A., Simmons, R., Fortin, M.-J., Day, C. C., & Landguth, E. L. (2023). Evaluating the evolutionary mechanisms maintaining alternative mating strategies in a simulated Bull Trout (Salvelinus confluentus) population. *Ecology and Evolution*, *13*, e9965. https://doi.org/10.1002/ece3.9965

Bernos, T. A., Day, C., Hill, J., Morissette, O., Jeffries, K. M., & Mandrak, N. E. (2023). Simulating the effects of long-distance dispersal and landscape heterogeneity on the eco-

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evolutionary outcomes of range expansion in an invasive riverine fish, Tench (Tinca tinca). *Molecular Ecology*, *32*(13), 3403–3418. https://doi.org/10.1111/mec.16964

Chang, W. (2025). *R6: Encapsulated classes with reference semantics*. https://doi.org/10.32614/cran.package.r6

Chang, W., Cheng, J., Allaire, J., Sievert, C., Schloerke, B., Aden-Buie, G., Xie, Y., Allen, J., McPherson, J., Dipert, A., & Borges, B. (2026). *Shiny: Web application framework for r*. https://CRAN.R-project.org/package=shiny

Day, Casey C., Alò, D., Simmons, R. K., Cotey, S. R., Zarn, K. E., Gazeley, I. F., Small, M., Fortin, M.-J., Bearlin, A. R., Smith, S. R., & others. (2024). Disentangling effects of dispersal, environment and anthropogenic barriers on functional connectivity in aquatic systems. *Molecular Ecology*, *33*(18), e17500. https://doi.org/10.1111/mec.17500

Day, Casey C., Landguth, E. L., Simmons, R. K., Baker, W. P., Whiteley, A. R., Lukacs, P. M., & Bearlin, A. (2020). Simulating effects of fitness and dispersal on the use of Trojan sex chromosomes for the management of invasive species. *Journal of Applied Ecology*, *57*(7), 1413–1425. https://doi.org/10.1111/1365-2664.13616

Day, Casey C., Landguth, E. L., Simmons, R. K., Baker, W. P., Whiteley, A. R., Lukacs, P. M., Davenport, K. A., & Bearlin, A. R. (2021). Evaluation of management factors affecting the relative success of a brook trout eradication program using YY male fish and electrofishing suppression. *Canadian Journal of Fisheries and Aquatic Sciences*, *78*(8), 1109–1119. https://doi.org/10.1139/cjfas-2020-0433

Day, Casey C., Landguth, E. L., Simmons, R. K., & Bearlin, A. R. (2023). CDMetaPOP 2: A multispecies, eco-evolutionary simulation framework for landscape demogenetics and connectivity. *Ecography*, *2023*(8), e06566. https://doi.org/10.1111/ecog.06566

Escalante, M. A., García-De León, F. J., Ruiz-Luna, A., Landguth, E. L., & Manel, S. (2018). The interplay of riverscape features and exotic introgression on the genetic structure of the Mexican golden trout (Oncorhynchus chrysogaster), a simulation approach. *Journal of Biogeography*, *45*, 1500–1514. https://doi.org/10.1111/jbi.13246

Forester, B. R., Day, C. C., Ruegg, K., & Landguth, E. L. (2023). Evolutionary potential mitigates extinction risk under climate change in the endangered southwestern willow flycatcher. *Journal of Heredity*, *114*(4), 341–353. https://doi.org/10.1093/jhered/esac067

Hijmans, R. J., Brown, A., & Barbosa, M. (2026). *Terra: Spatial data analysis*. https://CRAN.R-project.org/package=terra

Jombart, T., & Ahmed, I. (2011). Adegenet 1.3-1: New tools for the analysis of genome-wide SNP data. *Bioinformatics*. https://doi.org/10.1093/bioinformatics/btr521

Kamvar, Z. N., Tabima, J. F., & Grünwald, N. J. (2014). *Poppr*: An R package for genetic analysis of populations with clonal, partially clonal, and/or sexual reproduction. *PeerJ*, *2*, e281. https://doi.org/10.7717/peerj.281

Landguth, Erin L., Bearlin, A., Day, C. C., & Dunham, J. (2017). CDMetaPOP: An individual-based, eco-evolutionary model for spatially explicit simulation of landscape demogenetics. *Methods in Ecology and Evolution*, *8*(1), 4–11. https://doi.org/10.1111/2041-210X.12608

Landguth, Erin L., & Cushman, S. (2010). CDPOP: A spatially explicit cost distance population genetics program. *Molecular Ecology Resources*, *10*(1), 156–161. https://doi.org/10.1111/j.1755-0998.2009.02719.x

Landguth, Erin L., Day, C. C., Bearlin, A., Holden, Z. A., & Whiteley, A. R. (2018). Using simulation modeling to inform management of invasive species: A case study of eastern brook trout suppression and eradication. *Biological Conservation*, *221*, 10–22. https://doi.org/10.1016/j.biocon.2017.12.031

DRAFT

Landguth, Erin L., Forester, B., Shirk, A. J., Menon, M., Eckert, A. J., Whipple, A., Day, C. C., & Cushman, S. A. (2020). Modeling multilocus selection in an individual-based, spatially explicit landscape genetics framework. *Molecular Ecology Resources*, *20*, 605–615. https://doi.org/10.1111/1755-0998.13121

Landguth, Erin L., Holden, Z. A., Mahalovich, M. F., & Cushman, S. A. (2017). Using landscape genetics simulations for planting blister rust resistant whitebark pine in the US northern Rocky mountains. *Frontiers in Genetics*, *8*, 9. https://doi.org/10.3389/fgene.2017.00009

Landguth, Erin L., Williams, A., Roseman, M., Amadi, M., Fain, B., Kouete, M., Farrer, R. A., Haeseler, A. J., Razgour, O., Richardson, C., Weckworth, B., Weckworth, J., Whiting-Fawcett, F., Wilson, D., & Day, C. C. (2026). Simulating the landscape eco-evolution of host-pathogen systems with CDMetaPOP. *Ecological Informatics*, 103892. https://doi.org/10.1016/j.ecoinf.2026.103892

Lozada, A., Day, C. C., Landguth, E. L., & Bertin, A. (2023). Simulation-based insights into community uniqueness within fragmented landscapes. *Landscape Ecology*, *38*(10), 2533–2546. https://doi.org/10.1007/s10980-023-01735-6

Mendoza-Portillo, V., García-De León, F. J., & Der Heyden, S. von. (2023). Responses of population structure and genomic diversity to climate change and fishing pressure in a pelagic fish. *Global Change Biology*, *29*(14), 4107–4125. https://doi.org/10.1111/gcb.16732

Menon, M., Landguth, E. L., Leal-Saenz, A., Bagley, J. C., Schoettle, A. W., Wehenkel, C., Cushman, S., Waring, K. M., & Eckert, A. J. (2020). Tracing the footprints of a moving hybrid zone under a demographic history of speciation with gene flow. *Evolutionary Applications*, *13*, 195–209. https://doi.org/10.1111/eva.12795

Mims, M., Day, C. C., Hinkle, J., Fuller, M., Burkhart, J., Bearlin, A., DeHann, P., Holden, Z., & Landguth, E. L. (2019). Simulating demography, genetics, and spatially-explicit processes to inform reintroduction of a threatened char. *Ecosphere*. https://doi.org/10.1002/ecs2.2589

Nathan, L., Mamoozadeh, N., Tumas, H. R., Gunselman, S., Klass, K., Metcalfe, A., Edge, C., Waits, L., Spruell, P., Lowery, E., Connor, E., Bearlin, A., Fortin, M.-J., & Landguth, E. L. (2019). A simulation framework for evaluating fish hybridization dynamics in heterogeneous riverscapes. *Ecological Modelling*, *401*, 40–51. https://doi.org/10.1016/j.ecolmodel.2019.03.009

Savary, P., Foltête, J.-C., Moal, H., Vuidel, G., & Garnier, S. (2020). graph4lg: A package for constructing and analysing graphs for landscape genetics in r. *Methods in Ecology and Evolution*, *00*(1), 1–9. https://doi.org/10.1111/2041-210X.13530

Seaborn, T., Day, C. C., Galla, S. J., Höök, T. O., Jossie, E., Landguth, E. L., Liu, R., & Simmons, R. K. (2023). Individual-based models for incorporating landscape processes in the conservation and management of aquatic systems. *Current Landscape Ecology Reports*, *8*(3), 119–135. https://doi.org/10.1007/s40823-023-00089-8

Seaborn, T., Landguth, E. L., & Caudill, C. C. (2023). Simulating plasticity as a framework for understanding habitat selection and its role in adaptive capacity and extinction risk through an expansion of CDMetaPOP. *Molecular Ecology Resources*, *23*(6), 1458–1472. https://doi.org/10.1111/1755-0998.13774

Sullivan, S. L., Poorten, B. T. van, & Whitlock, M. C. (2026). An assessment of sampling strategies for surveying introgression in watersheds. *Canadian Journal of Fisheries and Aquatic Sciences*, *83*, 1–17. https://doi.org/10.1139/cjfas-2025-0255

van Etten, J. (2017). R package gdistance: Distances and routes on geographical grids. *Journal of Statistical Software*, *76*(13), 1–21. https://doi.org/10.18637/jss.v076.i13

Wickham, H. (2007). Reshaping data with the reshape package. *Journal of Statistical Software*, *21*(12), 1–20. https://www.jstatsoft.org/v21/i12/

DRAFT

Wickham, H. (2011). Testthat: Get started with testing. *The R Journal*, *3*, 5–10. https://doi.org/10.32614/RJ-2011-002

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. ISBN: 978-3-319-24277-4

Wickham, H., Danenberg, P., Csárdi, G., & Eugster, M. (2026). *roxygen2: In-line documentation for r*. https://doi.org/10.32614/cran.package.roxygen2

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2026). *Dplyr: A grammar of data manipulation*. https://CRAN.R-project.org/package=dplyr

Wickham, H., Vaughan, D., & Girlich, M. (2025). *Tidyr: Tidy messy data*. https://doi.org/10.32614/cran.package.tidyr

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